The Morin Z-Reduction Task (M-ZRT)

A constraint-based discriminative protocol for testing threshold models of positive affect access

Florian Morin
Independent Researcher in Affective Science
florianmorin.com
ORCID: 0009-0004-7301-8524

May 28, 2026

1 Operational definition (M-ZRT effect).

The M-ZRT effect is defined as a threshold-like, discontinuous transition into a distinct high positive affect regime, occurring under monitoring-disruption and action-continuation weakening conditions. A positive hit requires: (i) repeated clean failures prior to first access, (ii) abrupt onset (discontinuity rather than gradual improvement), (iii) qualitative distinctness from ordinary good mood or arousal, (iv) cross-context generalization beyond the task within 24 hours, and (v) at least one instance of persistence beyond acute stimulation when caffeine or other contextual modulators are present. This definition is intended to prevent post-hoc relabeling of ordinary enjoyment as a "threshold transition".

Within this framework, monitoring is not limited to explicit self-evaluation, but also includes the compulsory conversion of ongoing activity into optimized next-step selection, correction, anticipation, or continuation. The task therefore aims not only to reduce evaluative load, but to transiently weaken automatic action-continuation pressure without inducing global disengagement or sedation.

This protocol was initially derived from intensive self-experimentation conducted over extended daily sessions, and subsequently formalized into a constraint-based behavioral framework.

2 Aim

To test whether evaluative monitoring (Z) suppresses access to a distinct high positive affective regime ("ease"), producing a threshold-like pattern characterized by repeated failures followed by abrupt discontinuity. Monitoring is defined as any process involving evaluation, comparison to a standard, outcome anticipation, internal state checking, or compulsory conversion of ongoing activity into optimized next-step selection. The central claim is that access to high positive affect is not limited by insufficient activation, but by evaluative monitoring and action-continuation pressure acting as gating constraints. The task therefore aims not only to reduce evaluative load, but also to transiently weaken automatic correction, continuation, and optimization pressure during ongoing activity.

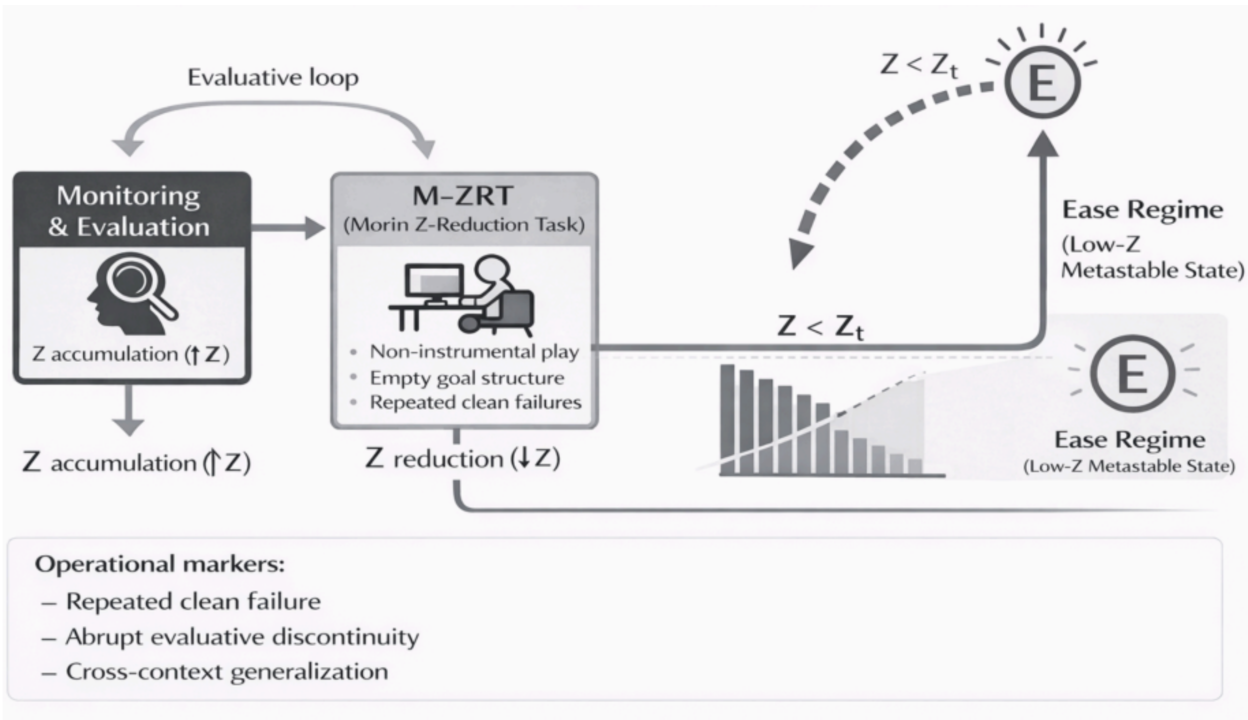

Figure 1: Schematic illustration of the M-ZRT framework. When Z falls below a threshold (Z < Zt), the system enters a low-Z metastable “ease” regime. Operational markers include repeated clean failures, abrupt discontinuity, and cross-context generalization.

Core manipulation

The manipulation is defined functionally rather than procedurally; specific implementations are interchangeable provided that they satisfy the same constraints. Low evaluative monitoring alone may be insufficient if activity remains strongly coupled to automatic continuation dynamics, compulsive progression, or persistent next-step recruitment.

Duration

7 days

Primary outcome (minimal log)

Daily access (0/1), latency (min), duration (min), intensity (0–10), and context.

Canonical signature (M-ZRT effect)

Repeated clean failures prior to first access, abrupt discontinuity, qualitative distinctness from ordinary mood or arousal, cross-context generalization within 24h, and persistence beyond acute stimulation (when contextual modulators such as caffeine are present).

Key falsifiers (non-negotiable)

1. Monitoring does not reduce success probability,
2. Contextual modulators alone reliably reproduce the effect without constraint conditions,
3. The effect correlates primarily with performance (e.g., winning or task success),
4. Reports are indistinguishable from ordinary arousal or mood improvement.

Operational threshold marker (optional)

For lightweight logs, a transition may be flagged when access = 1 and intensity ≥ 7. This serves as a standardized discontinuity marker rather than a full measurement model.

3 Alternative explanations and discriminative predictions

Several alternative explanations could account for reports of increased positive affect during the M-ZRT. The present framework does not assume exclusivity, but proposes discriminative predictions intended to separate competing interpretations.

Novelty explanation. One possibility is that observed effects reflect novelty rather than threshold access. Under a novelty account, transition probability should decrease monotonically with repeated exposure and should be strongly tied to unfamiliarity. In contrast, the threshold account predicts repeated clean failures followed by abrupt discontinuity, with possible persistence despite repeated exposure.

General arousal explanation. An alternative interpretation is that increased activation alone produces the reported effects. The present framework predicts that activation without monitoring reduction should be insufficient, and that contextual modulators alone should fail to reliably reproduce transitions. High arousal may even counteract access if it increases monitoring or optimization pressure.

Flow explanation. The proposed regime differs from classical flow accounts. Flow is generally associated with skill-demand matching, sustained task engagement, and performance coupling. The M-ZRT predicts reduced optimization, low instrumental stakes, weak dependence on task success, and possible occurrence outside performance contexts.

Absorption or dissociation explanation. Observed effects may reflect attentional absorption or transient dissociative processes. The threshold account predicts preserved environmental engagement and cross-context generalization rather than narrowing of awareness or detachment from surroundings.

Expectancy and placebo explanation. Participants may generate effects through expectation alone. The threshold account predicts weak temporal predictability, delayed first access, repeated failures prior to successful transitions, and reduced reproducibility with proceduralization.

Ordinary mood improvement explanation. The framework predicts qualitative discontinuity rather than gradual mood elevation. Abrupt onset, distinct phenomenology, persistence beyond immediate stimulation, and cross-context transfer are treated as critical markers separating threshold transitions from ordinary mood fluctuations.

Attentional capture explanation. Perceptual salience alone may account for positive effects. The present framework predicts that salience without monitoring reduction should preferentially produce activation or engagement rather than the proposed regime transition. Salience preservation is therefore treated as necessary but insufficient.

Descriptive dimensions of qualitative distinctness. Because the proposed regime is hypothesized to differ from ordinary positive mood or generalized arousal, qualitative distinctness should be described using multiple dimensions rather than a single intensity score. These dimensions are descriptive rather than constitutive and are not assumed to occur in every transition.

Potential dimensions include:

- **Abruptness of onset:** the transition is experienced as discontinuous or step-like rather than gradually increasing.

- **Sensory vividness:** increased perceptual richness, contrast, salience, or experiential intensity relative to baseline.
- **Reduction in comparison pressure:** temporary reduction in evaluative comparison, optimization, or pressure to determine whether the current state is sufficient.
- **Reduced next-step pressure:** diminished compulsion toward correction, continuation, planning, or immediate action selection.
- **Persistence beyond immediate stimulation:** the experiential shift partially remains after removal of the triggering context or contextual modulator.
- **Positive somatic component:** presence of localized bodily pleasure or unusually salient bodily positivity when present.
- **Cross-context transfer:** qualitative characteristics remain detectable across contexts rather than remaining tied to a single activity or stimulus.

These dimensions are intended as descriptive anchors to reduce post-hoc reinterpretation and improve discrimination from ordinary mood improvement, transient activation, or task engagement.

4 Evaluative load (Z) and salience capacity

Z denotes the level of evaluative monitoring and action-coupling pressure, including checking, control, outcome-oriented processing, anticipatory correction, and compulsory next-step selection. High Z constrains access by structuring experience around evaluation, continuation, and optimization. Reducing Z increases permissiveness, but only if salience capacity remains above a minimal threshold. Here, salience capacity refers to the ability of the system to sustain perceptual vividness, attentional engagement, and experiential differentiation. Reduced salience capacity, for example through fatigue, sedation, excessive underactivation, or flattening contextual factors, may shift the system toward weakly differentiated experiential states rather than the target regime.

In the present framework, action-continuation pressure is treated as a subcomponent of evaluative load (Z), rather than as an independent construct. Z does not refer only to explicit self-evaluation or reflective judgment, but to the broader degree to which ongoing experience is captured by correction, anticipation, comparison, and optimized next-step selection. Action-continuation pressure denotes the motor-cognitive form of this load: the tendency for the system to convert the present moment into a demand for continuation, adjustment, or improvement. It increases Z because it keeps activity organized around what should happen next. The M-ZRT therefore targets both explicit monitoring and this implicit continuation pressure, since either can preserve the evaluative structure that prevents threshold access

5 Methods

5.1 Pre-task Anti-Crystallization Orientation

Prior to task initiation, participants receive a brief orientation phase intended to reduce proceduralization and excessive evaluative framing. This phase is not designed as training and should not be repeated systematically across sessions.

Rather than learning a script or induction method, participants briefly familiarize themselves with common failure modes that may transform constraint-compatible perturbations into monitored procedures. Examples include optimization, repetition, effect-seeking, ritualization, performance checking, and active search for "good" implementations.

The objective is not to actively monitor these risks during task execution, but to recognize them if they become obvious and allow the procedure itself to recede into the background. Example implementations are provided in Appendix A. These examples are illustrative only and should not be interpreted as required components of the protocol.

Failure mode descriptions function as one-time anti-crystallization cues rather than monitoring tools. They should be read briefly, without memorization, rehearsal, or active recall during the session.

Participants may optionally expose themselves once to brief non-instrumental formulations intended to weaken habitual goal framing. These formulations are illustrative rather than procedural and should not be memorized, repeated, or treated as induction methods.

Examples include statements emphasizing the temporary suspension of optimization, production pressure, or immediate problem solving.

This orientation phase is not intended to induce a state or directly increase transition probability. Its function is limited to reducing procedural capture and preserving permissive conditions for spontaneous transitions.

6 M-ZRT Implementation Parameters

Implementation Notes and Contextual Modulation

Fast-paced retro-futuristic first-person shooters were selected because they satisfy multiple functional constraints hypothesized to support monitoring reduction while preserving salience. Older FPS environments are typically highly geometric, visually structured, and rich in contrast, providing strong perceptual capture without requiring extensive semantic interpretation or narrative engagement.

First-person perspectives may also reduce additional layers of representation and management associated with avatars or external camera control, producing a more direct perception-action loop. These games often minimize complex resource management, long-term planning, and multi-variable strategic optimization, thereby reducing opportunities for evaluative monitoring to re-emerge. Their continuous flow, frequent spatial changes, and high density of transient events, such as projectiles or rapid movements, may help maintain attentional capture while limiting explicit self-monitoring. Accessibility was also considered, as older FPS titles are often inexpensive or freely available and run on low-end hardware, facilitating replication across participants.

Importantly, no specific game or genre is theoretically required, these environments are presented only as implementations that satisfy the proposed functional constraints rather than as essential components of the mechanism. Their rapid and continuously shifting structure may transiently weaken deliberate action-selection loops by reducing the stability of ongoing correction and future-oriented control.

Low-dose caffeine was included as an optional contextual modulator under the assumption that threshold-like transitions may require preservation of salience in addition to reductions in evaluative load. Within a threshold framework, insufficient activation may prevent transition even when monitoring constraints are partially reduced. At minimal doses, caffeine may increase attentional engagement, sensory vividness, and general activation while remaining below the level at which strong arousal, explicit expectation, or self-monitoring become prominent. The emphasis on very small doses is intended to minimize the probability that caffeine itself becomes behaviorally salient, interpreted as an intervention, or incorporated into evaluative monitoring processes. Low-dose caffeine is not treated as increasing salience per se, but as preserving salience capacity under conditions where underactivation, fatigue, or perceptual flattening might otherwise reduce transition probability.

More generally, the M-ZRT can be implemented in any low-stakes environment satisfying a set of functional constraints: absence of performance feedback, absence of optimization goals, low instrumental salience, compatibility with attentional drift, partial suspension of compulsory continuation dynamics, and a monitoring stop rule.Any explicit performance-checking, feeling-checking, or deliberate attempt to determine “what should happen next” is theoretically inconsistent with the constraint structure and may substantially reduce interpretability.

Participants are encouraged to minimize strongly evaluative or cognitively demanding contexts on test days where feasible. Intrusive imagined social scenarios, such as anticipated arguments or confrontations, should also be reduced where possible.

More broadly, reducing exposure to news consumption, scrolling, comment reading, and social-media metrics may help limit re-engagement of evaluative monitoring. Similarly, reducing repeated time-checking and unnecessary muscular tension (e.g., jaw tension) may contribute to lowering baseline evaluative load. This includes avoiding a persistent goal-directed or “mission-oriented” mode of engagement, which tends to reinstate evaluative monitoring and compulsive action continuation, thereby increasing Z.

Repeated exposure to the protocol may decrease, rather than increase, the probability of entry. Sessions are therefore spaced by approximately 24 hours, not as a training schedule, but to reduce procedural carryover and allow monitoring processes to settle between attempts. Occasional omission of scheduled sessions is recommended to reduce temporal predictability and prevent expectancy formation.

The first successful transition may occur only after several null sessions. Within a threshold framework, such delay is not interpreted as gradual improvement, but as evidence of an entry barrier. Repeated failure followed by abrupt discontinuity is therefore treated as expected behavior rather than as an anomaly.

7 Example M-ZRT sequence with partial ordering constraints.

The following schematic illustrates an example implementation of the M-ZRT under partial ordering constraints.

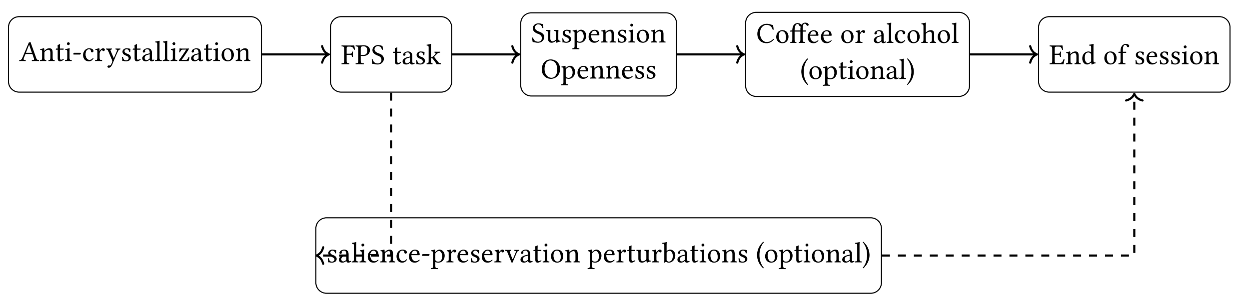

Figure 2: Example M-ZRT sequence with partial ordering constraints. Openness exercises are optional and may occur at variable points between FPS onset and session end.

The schematic below illustrates the two complementary conditions hypothesized to increase access to the ease regime. The relative contribution of salience-preserving conditions versus salience manipulations remains uncertain.

8 Non-monotonicity (U-shaped constraint)

Micro-perturbations are expected to follow a U-shaped profile: too little has no effect, too much becomes structured and reintroduces monitoring. For instance, a single micro-movement may be insufficient, whereas repeated movements (e.g., three or more) risk forming a pattern and becoming evaluable. Effective perturbations lie in a narrow intermediate range.

9 Task implementation example: HUD-off gameplay environment

Functional Constraints

Any admissible implementation must satisfy the following functional constraints:

1. **Low instrumental stakes.** The action must not carry meaningful cost, reward, or consequence for the participant. Once success matters, evaluative control is likely to re-engage.

2. **Brief interruption of compulsory continuation.** The task may introduce a short hesitation, deviation, incompletion, or unresolved transition, but not in a way that recruits sustained comparison, planning, or conscious problem-solving.

3. **No performance criterion.** There must be no internal standard of correct execution. If the participant begins to ask whether the procedure is being performed properly, the relevant condition is lost.

4. **No immediate self-evaluation.** The task must not be followed by checking, interpretation, or retrospective appraisal. Post hoc thoughts such as “was that right?” or “did it work?” may reintroduce monitoring processes and therefore reduce consistency with the proposed constraints.

5. **Minimal predictability without explicit surprise-seeking.** The implementation may contain mild arbitrariness or non-optimization, but it should not become a deliberate search for novelty, intensity, or effect.

6. **Non-rhythmic and non-ritualized execution.** If repeated, the action must not stabilize into a rhythm, routine, or familiar induction sequence. Ritualization converts perturbation into a monitored technique.

7. **Single-step permissiveness.** The participant should be able to perform the action and move on without correction, repetition, or refinement. Re-doing, improving, or adjusting the act reintroduces evaluative control.

8. **Non-dependence on task-specific content.** No specific game, object, movement, or setting is theoretically essential. Any implementation is only an example if it satisfies the same functional constraints.

9. **Absence of explicit expectation of effect.** The participant should not treat the action as a guaranteed trigger. Strong expectation, countdown logic, or “the effect should occur now” framing recruits anticipatory monitoring.

10. **Invalidity under monitoring capture.** Implementations that become heavily tracked, optimized, or proceduralized may progressively lose compatibility with the proposed constraint structure. Caffeine should be administered with variable timing, either prior to or following behavioral exercises.

9.1 Proceduralization Risks and Participant Instructions

Proceduralization represents a major theoretical failure mode of the M-ZRT. Because the framework assumes that evaluative monitoring and action-continuation pressure constrain transition probability, repeated use of identical sequences may progressively transform constraint-compatible perturbations into monitored procedures. Under this account, stabilization, optimization, or ritualization should reduce rather than improve transition probability.

Participants are therefore instructed to treat all examples as interchangeable implementations rather than techniques. Actions should not be repeated to improve performance, combined into fixed sequences, or interpreted as causal triggers. If participants begin asking whether an action was performed correctly, whether the procedure is working, or what should happen next, evaluative monitoring has likely re-emerged and the relevant condition is no longer satisfied.

To minimize procedural carryover:

- avoid repeating identical perturbation sequences across sessions,
- avoid counting, timing, or standardizing micro-perturbations,
- avoid explicit effect-seeking or transition anticipation,
- allow omission of sessions and variability in implementation,
- avoid retrospective checking immediately after perturbations,
- treat null sessions as theoretically informative rather than failed attempts.

The goal is not to perform the task correctly, but to preserve permissive conditions while minimizing the conversion of the protocol into a monitored skill.

10 Contextual Modulators: Monitoring Reduction vs Salience Preservation

Low-dose alcohol can be understood as a transient reducer of evaluative monitoring rather than a direct inducer of positive affect. At sufficiently low doses, it attenuates top-down control and self-evaluation, lowering Z while preserving experiential vividness. This allows mild mood elevation or subtle euphoria to emerge indirectly, through the relaxation of constraints rather than direct induction. The effect is strongly dose-dependent: beyond a narrow range, sedation and salience loss disrupt the permissive configuration.

Low-dose caffeine, by contrast, functions primarily as a salience-preserving or enhancing modulator. At minimal, non-salient doses, it increases attentional engagement and sensory vividness without substantially increasing monitoring. This can produce a mild sense of heightened presence, not as a direct mood effect, but by maintaining conditions favorable to spontaneous transitions. Higher doses tend to reintroduce monitoring through arousal and expectation, counteracting this balance. Contextual modulators alone are insufficient and act only under constraint conditions (see falsifiers).

Combining low-dose alcohol and caffeine may seem to jointly reduce monitoring and preserve salience, but it often reintroduces optimization. The pairing itself is easily interpreted as a deliberate strategy, shifting the system toward evaluation, timing, and dose-checking. This raises Z by triggering questions like whether the combination is working or optimal. As a result, two low-salience modulators can become a single monitored technique, undermining the permissive conditions they are meant to support.

Illustrative micro-perturbations (non-exhaustive, non-procedural)

The following examples are provided as illustrative micro-perturbations. They are not intended to be repeated, refined, or combined into a stable procedure.

Openness-oriented perturbations (monitoring attenuation)

- Ask a question internally and leave it unresolved. It can be an incomplete question, or just "what if?".
- Form a simple mental image (e.g., a red apple). Do not refresh it or attempt to sustain it. Simply notice when it fades.
- Generate a brief sense of recognition or approval without a target or recipient.

Salience-oriented perturbations (illustrative salience-preservation implementations)

- Select an object and label it as the most important in the game without looking at it directly. These perturbations are not hypothesized to directly induce transitions, but to help maintain minimal perceptual differentiation under permissive conditions. Salience-oriented perturbations may contribute less to transition probability itself than to perceptual richness, experiential differentiation, or hippocampal completion-like phenomena occurring during or after transition.

Suspension-oriented perturbations (brief disruption of ongoing control)

- While running, perform a brief micro-deviation without purpose, and immediately return to the activity without correction.

These examples are interchangeable. Any attempt to repeat, optimize, or stabilize them transforms the perturbation into a monitored procedure, thereby negating its function.

The schematic below illustrates the two complementary conditions hypothesized to increase access to the ease regime. "'latex

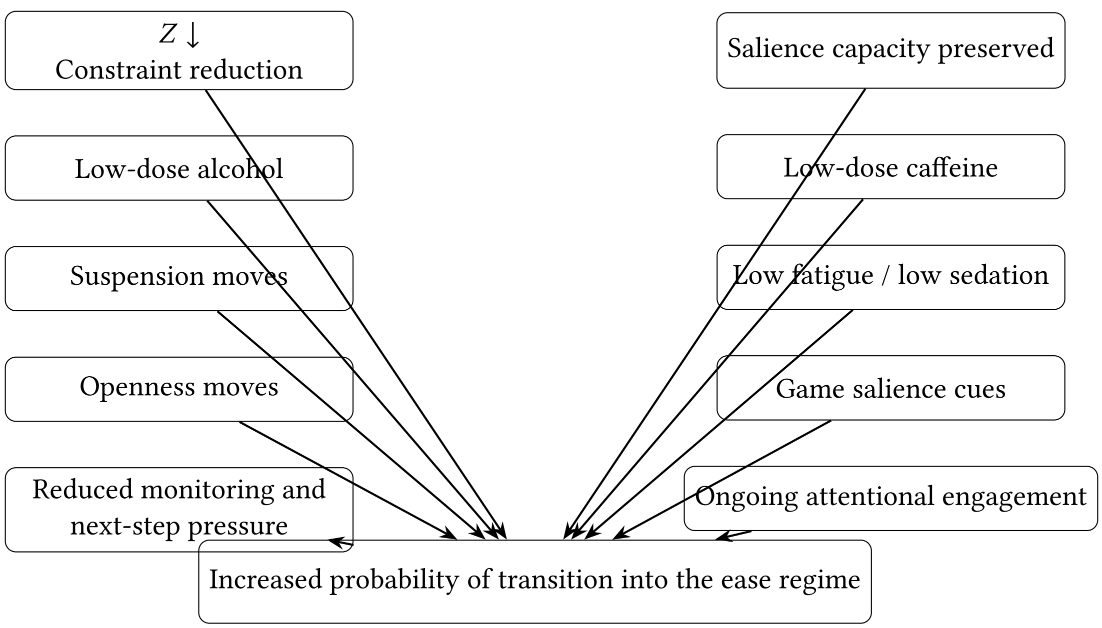

Figure 3: Illustrative schematic of two complementary conditions hypothesized to increase the probability of access to the ease regime: reduction in evaluative load ($Z\downarrow$) and preservation of sufficient salience capacity. Salience-oriented cues are not treated as direct inducers of transition, but as factors that help prevent experiential flattening under otherwise permissive conditions.

"'

The Morin Z-Reduction Task (M-ZRT): A constraint-based discriminative protocol for testing threshold models of positive affect access

11 Minimal empirical predictions

The framework generates several near-term predictions that can be tested using lightweight designs:

- repeated failures should precede successful transitions more frequently than expected under monotonic learning accounts,
- first successful transitions should cluster discontinuously rather than improve gradually across sessions,
- contextual modulators alone should produce weaker effects than constraint-compatible implementations,
- performance metrics should weakly predict access probability,
- proceduralization should reduce rather than improve transition probability.

12 Standardized test conditions (minimal)

To reduce interpretive ambiguity while keeping the protocol lightweight, each session should be tagged using the following minimal condition grid. This does not aim to fully control the environment, but to prevent "same protocol" claims under substantially different contexts. These tags are recorded post hoc and should not be actively tracked or monitored during the session.

Variable	**Levels (record one)**	**Notes**
Fatigue state	Low / Moderate / High	Sessions under "High" fatigue are expected to show reduced entry probability and may be postponed.
Caffeine	None / Low-dose / Moderate+	Low-dose is treated as a contextual modulator. Avoid moderate+ due to arousal confounds.
Alcohol	None / Low-dose / Moderate+	If used, restrict to very low-dose only. Moderate or hi gh doses substantially increase interpretive ambiguity due to arousal or intoxication confounds.

Optional Additional Tags

These tags are not required but may help clarify failure modes without increasing measurement burden.

Variable	Levels
Sleep debt	None / Mild / Significant
Current load	Low / Moderate / High (work, obligations, unresolved tasks)
Rumination	Low / Moderate / High

13 Quantitative and distributional predictions

Beyond qualitative signatures, the threshold framework generates distributional predictions that differ from monotonic learning accounts.

Prediction 1: High early-session variability. Transition probability should show substantial between-session variability during early exposure. Participants may exhibit repeated null sessions followed by isolated successful transitions rather than smooth improvement curves. Consequently, variance should remain elevated during initial sessions rather than progressively narrowing.

Prediction 2: Clustering of first successful transitions. If access depends on crossing a threshold rather than cumulative skill acquisition, first successful transitions should cluster discontinuously. The distribution of first-access events should therefore show concentration around particular sessions rather than approximately linear improvement across days.

Prediction 3: Weak monotonic improvement. Learning-based accounts predict progressive increases in success probability across repeated exposure. The threshold model instead predicts weak or absent monotonic trends, with abrupt transitions dominating over smooth performance gains.

Prediction 4: U-shaped perturbation function. Micro-perturbations are predicted to exhibit a non-monotonic relationship with transition probability. Insufficient perturbation should produce little effect, whereas excessive perturbation should increase monitoring and reduce success probability. Intermediate perturbation magnitude should therefore maximize access probability.

Prediction 5: Reduced performance coupling. If the proposed regime differs from conventional task engagement states, objective task performance should weakly predict transition probability. Winning, score, or efficiency should explain less variance than monitoring-compatible conditions.

Prediction 6: Proceduralization cost. Repeated stabilization of the same sequence of actions should reduce transition probability over time. Thus, increasing procedural regularity should predict lower rather than higher success rates.

Prediction 7: Heavy-tailed access distribution. Transition frequency is predicted to be unevenly distributed across sessions and participants. A minority of sessions may produce disproportionately many successful transitions, whereas many sessions may remain null despite similar conditions.

These predictions are intended to increase falsifiability by shifting evaluation from isolated reports toward measurable distributional signatures.

A Illustrative Micro-Perturbations and Failure Modes

A.1 Question Without Answer

Description. A question briefly appears internally (e.g., "What if I turned left?"), but no attempt is made to resolve, elaborate, or answer it. Activity continues normally.

Why this example may satisfy task constraints. Incomplete questions may transiently weaken compulsory closure and immediate next-step recruitment without requiring explicit decision-making or prolonged attentional engagement. Because the question remains unresolved, it may briefly disrupt automatic continuation dynamics while preserving ongoing activity.

Potential failure modes.

- completion pressure through automatic answer generation,
- route simulation, prediction, or pros/cons analysis,
- meta-monitoring ("Did I avoid answering correctly?"),
- repeated generation of questions becoming a search task,
- ritualization through reuse of identical question formats,
- persistence of unresolved questions occupying working memory,
- transformation into an optimization strategy,
- deliberate search for more effective questions,
- accumulation of unfinished loops across repetitions,
- explicit expectation that the question should produce openness or reduce Z.

Because incomplete questions can automatically recruit prediction and closure processes, repeated or intentional use may progressively transform the perturbation into a monitored procedure.